

Supplementary Information

Stress transfer and dislocation response near a constrained $\text{Al}_{13}\text{Fe}_4$ inclusion in aluminium

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S1 Structure and preparation

The interface cell contains 91 428 atoms. Matrix atoms have identifiers 1–72 532; the remaining 18 896 belong to the inclusion. Elemental types are Al, Fe, Mg and Si in that order in the LAMMPS input mapping, although the interface configurations contain no Mg or Si. There are 87 076 Al and 4352 Fe atoms in the complete cell. The fcc aluminium axes are $[\bar{1}10]$, $[11\bar{2}]$ and $[111]$, with lattice parameter 4.05 Å. The $\text{Al}_{13}\text{Fe}_4$ cell is monoclinic [1]; the same Jelinek interaction potential is used across the interface and within each phase [2].

The initial box has $L_x = 154.644253$ Å, $L_y = 64.482817$ Å and z limits -10.0 and 165.943041 Å. The matrix construction uses 56 (111) layers. Its upper material surface differs from the box boundary and moves during preparation, so a single exact free-surface height is not assumed throughout. The support layer extends to $z = 20$ Å. The ridge is centred at $x = 77.3221$ Å and has semi-axes 35 and 20 Å in the xz plane.

The inclusion is scaled in-plane by factors 0.9978336 and 0.9973955 to match the periodic cell. Matrix atoms closer than 2.2 Å to inclusion atoms are removed during construction. The resulting minimum separation is approximately 2.219 Å within the cell and 2.292 Å across the periodic seam. These geometric checks exclude initial overlaps at that distance scale; they do not validate the relaxed interface energy or structure. The lower region $z < 6$ Å is fixed.

LAMMPS uses metal units, an atomic particle style and boundaries periodic in x and y and fixed in z [3]. The neighbour skin is 2 Å. For minimisation, neighbours are rebuilt at every step. The prepared control undergoes conjugate-gradient minimisation with requested force two-norm 1 eV/Å, followed by 6 ps at 300 K using a Nosé–Hoover damping time of 0.1 ps. Langevin cooling then lowers the target temperature to 0.5 K over 12 ps, with damping time 0.2 ps. The velocity and cooling seeds are 4242 and 4243. Velocities are zeroed before a final minimisation whose requested force two-norm is 0.05 eV/Å. A reflecting wall at the upper box edge is present during the thermal preparation. This preparation defines the starting state of the paired static calculation.

Both members of the static pair are reconstructed from that same control snapshot. The 4816 inclusion atoms above $z = 20$ Å form the displaced ridge subset. Their recorded centroid is approximately (77.258, 32.201, 28.678) Å. The perturbation tensor is

$$\mathbf{E}_0 = \begin{pmatrix} 0.000485 & 0 & 0.001455 \\ 0 & -0.000970 & 0 \\ 0.001455 & 0 & 0.000485 \end{pmatrix}. \quad (\text{S1})$$

A reference coordinate changes by $\delta \mathbf{r} = \mathbf{E}_0(\mathbf{r} - \mathbf{r}_c)$. The largest imposed displacement is 0.05412 Å. The determinant of $\mathbf{I} + \mathbf{E}_0$ is $(1 + \varepsilon_0)(1 - \varepsilon_0/2)^2$, with $\varepsilon_0 = 0.00194$. Its depar-

ture from unity is about 2.82×10^{-6} , consistent with volume conservation only to first order in strain.

All inclusion atoms are restrained by

$$U_s = \frac{K}{2} \sum_{i \in \text{inclusion}} |\mathbf{r}_i - \mathbf{r}_i^{\text{ref}}|^2, \quad K = 20 \text{ eV/\AA}^2. \quad (\text{S2})$$

The support references are unshifted in both copies; the ridge references differ by Eq. (S1). The spring energy is included in minimisation. Conjugate gradients use maximum displacement $0.05 \text{ \AA}$, zero energy tolerance, force two-norm tolerance $0.02 \text{ eV/\AA}$, at most 6000 iterations and 60 000 force evaluations. The held control stops on force tolerance after 235 iterations, with final force two-norm $0.019779915 \text{ eV/\AA}$. The held perturbed state also stops on force tolerance, after 310 iterations at $0.019879873 \text{ eV/\AA}$. These values verify the stated stopping criterion. They do not establish convergence of a stress profile with respect to force tolerance, geometry or starting state.

S2 Stress evaluation and distances

The stored per-atom quantities are the six components from `compute stress/atom NULL virial`, in the order xx, yy, zz, xy, xz, yz . For a control-selected set of N Al atoms, the analysis conversion is

$$\sigma_{\alpha\beta} = -\frac{1}{10N\Omega} \sum_{i=1}^N S_{i,\alpha\beta}, \quad \Omega = 16.6072 \text{ \AA}^3, \quad (\text{S3})$$

where the stored S_i are in bar $\text{\AA}^3$ and the result is in MPa. This conversion makes normal compression positive and gives the negative of the tensile-positive Cauchy tensor. Signed MD quantities in the manuscript retain this pressure convention. The continuum comparison uses absolute shear projections, which are unaffected by a global sign change. No kinetic term is included. This configurational measure excludes an interpretation of the profile as a directly measured temperature or heat flux.

Atom identities and element types are matched between snapshots. The control coordinates determine membership in each layer and in the $20 \text{ \AA}$ axial window. The window centre is obtained from the midpoint of the control atom x extents. Layers contain type-1 atoms and span $4 \text{ \AA}$ in z . A layer is retained only when its lower edge lies above every inclusion atom in both snapshots. Inclusion IDs exceed 72 532 and comprise both Al and Fe. Their maximum heights are $39.707546 \text{ \AA}$ in the control and $39.718591 \text{ \AA}$ in the perturbed state. The 23 retained centres run from $r = 22$ to $110 \text{ \AA}$, where $r = z - 20 \text{ \AA}$. The last layer edges are $r = 108$ and $112 \text{ \AA}$. This choice omits layers containing inclusion aluminium and the uppermost free-surface layers.

The tensor average is formed separately for each member of the pair and then subtracted. With $\mathbf{D} = \Delta\boldsymbol{\sigma} - \text{tr}(\Delta\boldsymbol{\sigma})\mathbf{I}/3$, the von Mises measure is $\sqrt{3\mathbf{D} : \mathbf{D}}/2$. The largest absolute slip-system projection is $\max_{\alpha} |\mathbf{s}_{\alpha} \cdot \Delta\boldsymbol{\sigma} \mathbf{n}_{\alpha}|$. The xz projection and its sign are preserved independently. Consequently, a small width-averaged shear can coexist with a larger axial-window shear of the opposite sign.

The far-layer statistics are evaluated over thirteen retained layers with $r > 60 \text{ \AA}$. The mean maximum projection is 2.477 MPa and its spatial standard deviation is 0.543 MPa . The bins are not independent replicas of a homogeneous state. These statistics quantify spatial variation and cannot be used as standard errors, numerical noise estimates or a physical detection threshold. Residual stress at the largest sampled distance is reported explicitly.

An attenuation height can be defined only relative to a stated stress level. If $h_m(s)$ denotes the height above the nominal crest at which the descending axial-window magnitude reaches s , linear interpolation between adjacent retained bins gives approximately 4.69 , 6.69 and 8.55 nm

for $s = 10, 5$ and 2 MPa, respectively. These are operational profile descriptors, with 0.4 nm sampling, rather than independently determined material lengths. No zero-stress attenuation height is inferred.

S3 Deformation retention and free diagnostics

A least-squares affine map relates the positions of corresponding Fe atoms in the two minimised configurations. After removing translations, the fit gives a deformation gradient $\mathbf{F}$. Minimum-image displacements are used along the periodic directions. The retained strain is represented by

$$\mathbf{E}_{\text{res}} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{I}), \quad \eta = \frac{\mathbf{E}_{\text{res}} : \mathbf{E}_0}{\mathbf{E}_0 : \mathbf{E}_0}. \quad (\text{S4})$$

For the 1008 Fe atoms with control $z > 22$ Å, $\eta = 0.9728$. The principal reported components are $E_{xx} = 0.000482$, $E_{yy} = -0.000887$, $E_{zz} = 0.000460$ and $E_{xz} = 0.001435$. The fit residual has root-mean-square magnitude 0.0035 Å.

The formal standard error 0.0038 propagates the least-squares coefficient covariance, using pooled residual variance over $3N - 12$ degrees of freedom. The atomic residuals are spatially correlated; this fit statistic does not include model error or variation between independent trajectories. A scalar projection also does not guarantee that every tensor component retains the same fraction of the imposed deformation.

Unrestrained minimisations are retained only as diagnostics. For the perturbation pair without springs, the ridge projection is 0.4364 ± 0.1019 in the saved output. Both minimisations stop at 6000 iterations with final force two-norms 0.8660473 and 0.48161949 eV/Å for control and perturbed states, respectively, above the requested 0.02 eV/Å. A separate direct minimisation record gives 0.2007 ± 0.1064 for a differently prepared pair. Neither record establishes an equilibrium retained fraction, a relaxation time or a field after magnetic removal. Their stress profiles are excluded from the main quantitative comparison.

S4 Loading and line tracking

The loaded interface geometry retains 91 428 atoms and has the same orientation and inclusion shape as the static cell. The initially inserted lower and upper edge lines are centred at approximately $(139.322, 49.228)$ and $(115.922, 72.611)$ Å in the xz plane. Their displacements are generated by a periodic Volterra construction. These insertion coordinates differ from the coordinates after minimisation and initial dynamics.

The loading input first minimises the constructed cell with a requested force two-norm of 1 eV/Å and a 2000-iteration limit. It then assigns velocities at 300 K using seed 88004 and integrates with a 1 fs step. The bottom region is fixed, mobile atoms are coupled to a Nosé–Hoover thermostat with damping time 0.1 ps, and a reflecting wall is placed at the upper box boundary. Equal forces along x on the top selected atoms transmit the nominal shear traction. The inclusion is either unrestrained or attached to springs of 20 eV/Å², as specified by the case.

Let s be the number of steps after the 5000-step preload, $T_s = 16\,000$ steps and N the total run length. The nominal stress in MPa is

$$\tau(s) = \begin{cases} 0, & s < 0, \\ q \left[\frac{s}{2} - \frac{T_s}{2\pi} \sin\left(\frac{\pi s}{T_s}\right) \right], & 0 \leq s < T_s, \\ q(s - T_s/2), & s \geq T_s, \end{cases} \quad q = \frac{\tau_{\text{max}}}{N - 5000 - T_s/2}. \quad (\text{S5})$$

The unrestrained control uses $N = 101\,000$ and $\tau_{\text{max}} = 400$ MPa. The restrained control and perturbed runs use $N = 45\,000$ and $\tau_{\text{max}} = 145.45$ MPa. Their post-onset rates agree to the

precision of these rounded inputs. Regular snapshots are saved every 2000 steps. The analysis tables use the actual endpoint for each run, giving 9.0909 MPa between late frames in the free control and 9.090625 MPa in the restrained pair.

DXA identifies Burgers vectors and line segments [4, 5]. Line positions are unwrapped in x using $L_x = 154.644253 \text{ \AA}$. For each Burgers-vector family, a linear baseline is fitted over steps at or above 4000 with nominal stress below 40 MPa. The departure criterion is a residual magnitude greater than $\max(6s_x, 8 \text{ \AA})$, where s_x is the baseline residual standard deviation. The subsequent two sampled positions are checked where available; a missing line is accepted by the stored detector. The numerical criterion therefore requires a separate continuity check near boundaries and after segment disappearance.

The upper-line baseline residual standard deviations are 3.8919 and 3.5046 $\text{\AA}$ in the restrained control and perturbed runs. The corresponding thresholds are 23.351 and 21.027 $\text{\AA}$. The first accepted samples are at 38 and 40 ps, respectively. The same criterion consequently uses different fitted baselines and displacement thresholds. Its one-frame difference cannot be interpreted as an independently measured 9.09 MPa strengthening increment.

The lower family disappears by 10 ps in both restrained runs. Its later family-labelled positions can belong to seam segments and do not provide a valid second onset for the original line. At the last analysed control frame, 44 ps, the total extracted length is 39.344 $\text{\AA}$ in three segments. The family-aggregated lengths are 26.191 and 13.153 $\text{\AA}$; these are not necessarily lengths of individual connected lines. No segment is extracted from the corresponding perturbed frame. The results distinguish absence of a detected line from proof of an atomistic reaction pathway.

For the unrestrained control the lower-line departure is at the 36 ps sample, 104.545 MPa, preceded by 95.455 MPa. The upper-line criterion is met at 40 ps, 122.727 MPa. The final regular analysis frame is 100 ps, 395.455 MPa; the nominal 400 MPa endpoint is not covered by that table. Short seam signals occur transiently around 232–241 MPa and again above 377 MPa. They do not establish persistent interface nucleation. A scalar threshold exceeding 395.455 MPa is not inferred from the absence of a confirmed event.

S5 Separate alloy comparator

The alloy cell contains 92 800 atoms with dimensions $114.5513 \times 49.6022 \text{ \AA}$ in-plane and box limits $z = -8.0$ and $283.2392 \text{ \AA}$. Its counts are 91 315 Al, 928 Mg and 557 Si atoms. The nominal whole-cell concentrations are therefore 1.0 at.% Mg and approximately 0.6 at.% Si. Substitution is restricted to the mobile region; the boundary layers remain aluminium. The bottom fixed layer is 8 $\text{\AA}$ thick and the top loaded layer 12 $\text{\AA}$. The edge partners are separated by 205.768 $\text{\AA}$ along the interface normal.

The initial minimisation stopped on energy tolerance with force two-norm 10.118429 eV/ $\text{\AA}$ and maximum force component 1.9509711 eV/ $\text{\AA}$. It did not reach the requested force tolerance. The subsequent 300 K trajectory was completed, but this preparation does not demonstrate mechanical equilibration of the initial dislocation arrangement. The trajectory has 10 ps without applied shear and successive nominal levels of 45, 55, 65 and 75 MPa, each held for 30 ps. Each increase is smoothed over 4 ps. The recorded net displacement of the lower line is 1.98 $\text{\AA}$ from 10 to 130 ps, with a position standard deviation of 0.63 $\text{\AA}$. The preceding zero-load interval is excluded; the displacement across the complete 0–130 ps trajectory is 6.99 $\text{\AA}$. The per-level net displacements are 0.58, 1.56, -1.53 and 0.17 $\text{\AA}$. The net motion over the settled portion of each level is less than one Burgers-vector magnitude in this one solute realisation.

These observations are conditional on the sampled solute arrangement, internal stress and short observation time. The applied stress is not a measurement of the local resolved traction on the dislocation. The trajectories therefore support a statement of limited motion under the applied programme, without establishing an intrinsic 75 MPa obstacle strength. No activation volume is extracted from the fitted slopes.

The public comparator package contains the complete saved atomic trajectory, including its original coordinate and stress columns, in compressed shards under `data/publication/alloy/trajectory.*.lammprj.gz`. The derived 131-frame position table, summaries, initial and final structures and log are also provided. The available initial file was saved after the recorded run date. Its atom identities, types and fixed boundary coordinates agree with the trajectory, but no launch-time hash or original copy certifies its mobile coordinates. It is therefore a candidate starting structure, not a certified exact replay input. The saved trajectory allows independent DXA reconstruction and reassessment of the spatial selection used in the comparator analysis. An activation volume remains unidentified by these observations.

The original quiet-region stress selection included Al and Si while excluding Mg (type 2). Reanalysis of all 131 saved frames reproduces the archived displacements and stresses. Selecting only Al or all alloy atoms changes the mean stress on a loading level by less than 1.71 MPa. The corresponding diagnostic estimates of V^*/b^3 are -12.32 ± 21.31 for the original selection, -13.54 ± 19.19 for Al and -12.32 ± 19.83 for all atoms in the pressure-sign convention. Using tensile-positive shear reverses these fitted signs. These fits to fluctuating positions do not determine a physical activation volume. The comparison is recorded in `stageG6_stress_selection_rela.json`.

S6 Continuum and activated-glide estimates

The sphere reference has the same small-strain tensor as Eq. (S1), radius 35 Å, shear modulus 26.5 GPa and Poisson ratio 0.347. The inclusion and unbounded matrix have identical isotropic stiffness. For the Eshelby tensor $\mathbf{S}$ and stiffness $\mathbf{C}$, the interior stress is

$$\boldsymbol{\sigma}_{\text{in}} = \mathbf{C} : (\mathbf{S} : \mathbf{E}_0 - \mathbf{E}_0). \quad (\text{S6})$$

The maximum interior slip-system projection is 41.45 MPa [6]. This is an interior value for a transformation-strain problem. It is not the surface amplitude of every exterior component. The ridge in the main calculation has different constraints, a support and a nearby free surface, so the comparison is not an ordering theorem or a validation of the atomistic boundary condition.

The scalar creep calculation assumes a compact-particle exterior amplitude $\tau_m(a/R)^3$ and volume fraction f . To leading order in f , the spatial weight of stress magnitudes between s and $s + ds$ is $f\tau_m s^{-2}ds$ for $0 < s < \tau_m$. This weight is used only under an integral for the excess rate. It is not a normalised probability density extended to zero stress. Equal positive and negative stress weights and a stress-independent activation volume give

$$H = fx \int_0^x \frac{\cosh u - 1}{u^2} du = f\{x \text{Shi}(x) - [\cosh(x) - 1]\}, \quad x = \frac{V^*\tau_m}{k_B T}. \quad (\text{S7})$$

Here $\text{Shi}(x) = \int_0^x \sinh(u)/u du$. The integrand has the finite limit 1/2 at the origin. Particle overlap, grain-orientation distributions and stress-dependent barrier shapes are omitted.

For the numerical sensitivity calculation, $k_B T = 4.141947 \times 10^{-21}$ J at 300 K, $b = 2.86378246$ Å and $f = 0.00246$. The volume fraction is a modelling scenario. For a mass fraction $w = 0.0035$, densities $\rho_i = 3.85$ and $\rho_m = 2.70$ g/cm³ give

$$f = \frac{w/\rho_i}{w/\rho_i + (1-w)/\rho_m} \simeq 0.00246. \quad (\text{S8})$$

These assumed densities and composition do not establish the volume fraction of every experimental specimen. The activation-volume range $19b^3$ – $142b^3$ is a sensitivity interval. The reported value $70b^3$ for an Al–Mg–Si alloy processed by equal-channel angular extrusion provides a comparison from a different material state [7].

At $H = 0.25$, Eq. (S7) gives $x = 6.754241794$. The required amplitudes are 62.691, 39.705, 23.823, 17.016, 11.911 and 8.388 MPa for $V^*/b^3 = 19, 30, 50, 70, 100$ and 142, respectively. Conversely, $\tau_m = 15.451$ and 41.45 MPa require $V^*/b^3 = 77.091$ and 28.737. The first amplitude is an axial-window ridge maximum; the second is the sphere interior projection. Their substitution into the scalar exterior model is explicit and conditional. It does not calibrate magnetostriction or demonstrate the experimental creep rate.

S7 Records and reproducibility

The publication data and code are located at <https://github.com/iluuua/magnetostriction-plasticity-bounds>. The manifest under `data/publication` records the source configurations, retained columns and checksums. It distinguishes the complete atomic-virial snapshots of the static pair, the full-column alloy trajectory and position-only projections of the three interface loading trajectories. The interface projections preserve particle identifiers, types, coordinates, frame times and cell bounds for every saved regular frame. These projections do not provide all original per-atom stress and energy columns.

The stress results derive from `stageG10_field_profile.json` and its associated CSV, calculated from the held control and perturbed atomic-virial snapshots. The layer selection, signed projections and reported peaks can therefore be recomputed from the primary static data. The deformation-retention components and formal fit error are in `stageG12_eigenstrain_retention.json`. The diagnostic free outputs are identified by `stageG12_eigenstrain_retention_free.json` and `stageG12_eigenstrain_retention_free_cg.json`; their lack of convergence limits their use as set out in Section S3.

Interface motion is recorded in `stageG2_depinning_summary_G15held.json` and `stageG2_depinning_summary_G15ctl_free.json`. The associated tables `stageG2_depinning_G15_ctl_G15held.csv`, `stageG2_depinning_G15_fld_G15held.csv` and `stageG2_depinning_G15_ctl_G15ctl_free.csv` contain frame times, nominal stresses and family-resolved positions and lengths. The published coordinate trajectories allow independent reconstruction of line content. The detector summaries require the continuity qualification in Section S4; later entries for a family need not represent the same original line.

The alloy comparator uses `stageG6_vstar_rela_frames.csv` and `stageG6_vstar_rela_summary.json`. Only the position history and the completed loading range are used here. The fitted activation volume stored in a diagnostic summary is not treated as a physical estimate. The sphere interior comes from `stageG8_eshelby3d.json`. The integral and inverse-stress results are defined by Eq. (S7), with numerical inputs specified above; the corresponding archived calculation is `stageG5_two_scale_bridge.json`. Narrative fields in records do not substitute for these definitions or for convergence evidence.

The figure-generation code operates on these configurations and records. Model renderings use OVITO and represent simulated atomic positions. No simulated image is presented as an experimental micrograph. The four main figures contain the geometry, static field, interface trajectories and conditional activated-glide dependence. Figure labels follow the language of each manuscript; the plotted numerical data are identical.

References

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